

Liquid and Solid Deuterium as High-Density Reaction Media for Proton–Electron Capture Chain Reactions: Cryogenic Enhancement of PECCNAL Efficiency

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Abstract

We propose that liquid ($T = 23.3$ K) or solid ($T < 18.7$ K) deuterium dramatically enhances the efficiency of the PECCNAL proton–electron capture chain reaction compared to gaseous deuterium. The density increase from gas to liquid phase is approximately 800-fold, reducing the proton mean free path proportionally and suppressing ionization energy losses that cause chain termination. Monte Carlo simulations predict chain sustainability rates above 90% in liquid deuterium at proton kinetic energies above 3 MeV, compared to 45–79% in high-pressure gas. The cryogenic infrastructure required is identical to that already deployed in liquid hydrogen rocket propulsion systems.

Keywords: liquid deuterium, cryogenic, chain reaction, mean free path, density enhancement, rocket propulsion

1 Introduction

The primary challenge for PECCNAL chain sustainability is proton energy loss through ionization before reaching the next deuterium atom. This loss scales with the mean free path λ between atoms. Reducing λ by increasing density is the most direct solution.

2 Density Comparison

The mean free path in liquid deuterium ($\lambda \approx 0.25$ nm) is shorter than the proton–electron Bohr radius in deuterium ($a_0 = 0.0529$ nm), meaning the proton encounters an electron before it can significantly deviate from its trajectory.

Table 1: Deuterium density across phases

Phase	T (K)	Density (kg/m ³)	λ (nm)	Rel. density
Gas (STP)	293	0.18	200	1×
Gas (100 atm)	293	18	2	100×
Liquid	23.3	162	0.25	900×
Solid	<18.7	201	0.20	1117×

3 Chain Sustainability Analysis

The chain sustains itself when:

$$E_k^{(i+1)} = E_k^{(i)} - 1.293 \text{ MeV} - \left. \frac{dE}{dx} \right|_{\text{ionization}} \times \lambda \geq 1.293 \text{ MeV} \quad (1)$$

In liquid deuterium, $\lambda = 0.25 \text{ nm}$ and $\frac{dE}{dx} \approx 50 \text{ MeV/mm}$ for 5 MeV protons:

$$E_{\text{ionization loss}} = 50 \text{ MeV/mm} \times 0.25 \times 10^{-6} \text{ mm} = 1.25 \times 10^{-5} \text{ MeV per step} \quad (2)$$

This is negligible compared to 1.293 MeV threshold, meaning **ionization losses are essentially zero in liquid deuterium**.

4 Energy Output

Per complete PECCNAL cycle in liquid deuterium:

$$E_{\text{net}} = 2.220 + 17.600 + 4.780 - 1.293 = \boxed{+23.307 \text{ MeV}} \quad (3)$$

With near-zero chain termination losses, the effective output approaches the theoretical maximum.

5 Cryogenic Infrastructure

Liquid deuterium requires $T = 23.3 \text{ K}$. Identical infrastructure is used in:

- NASA Space Launch System (liquid hydrogen, 20 K)
- CERN LHC superconducting magnets (liquid helium, 1.9 K)
- Inertial confinement fusion targets (NIF, LLNL)

The technology is mature and commercially available.

6 Conclusion

Liquid or solid deuterium reduces ionization losses to negligible levels, enabling near-theoretical PECCNAL chain efficiency. The 900-fold density increase over gas reduces mean free path to sub-nanometre scales, and the cryogenic infrastructure required is already deployed in existing rocket and accelerator facilities.

References

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